

REPORTE DE CONFIGURACIÓN Y VALIDACIÓN DE QoS EN GNS3

Datos generales

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- **Materia:** Seguridad en Redes de computo
- **Práctica:** Implementación de QoS en GNS3
- **Fecha:** 29 de abril 2026
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Objetivo

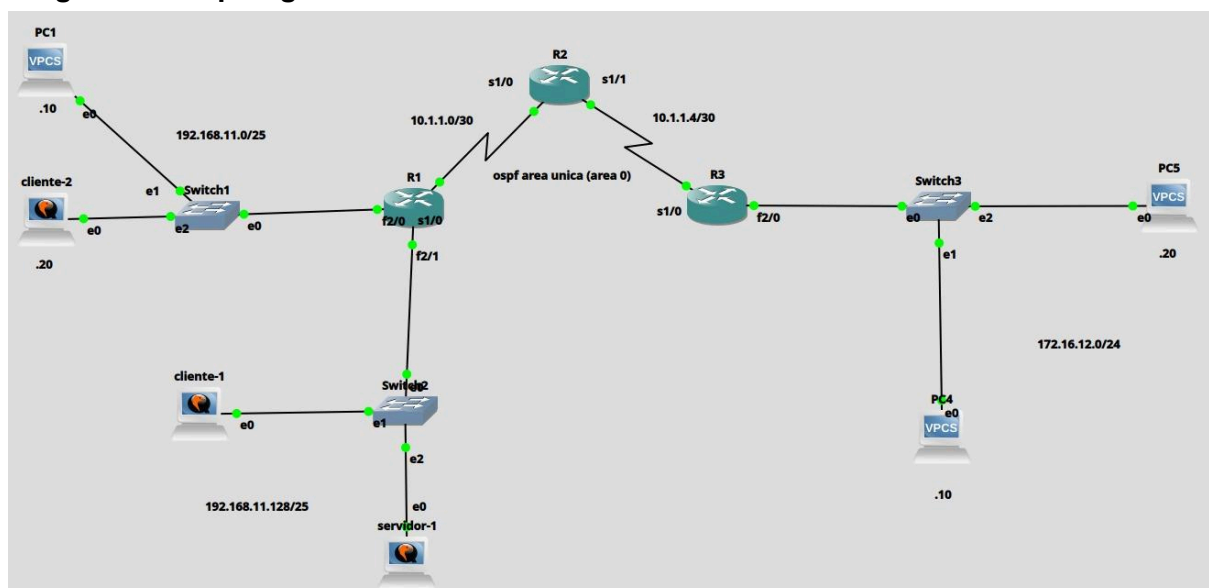
Implementar y validar políticas de Calidad de Servicio (QoS) en una topología de GNS3, utilizando diferentes criterios de clasificación, marcación de paquetes (IPP y DSCP) y verificación mediante captura de tráfico.

1. Topología de red

Descripción breve:

Se implementó una red en GNS3 utilizando routers c7200, un switch Ethernet, VPCS y máquinas virtuales con Debian 13.

Imagen de la topología:



2. Exploración del router

Modelo: c7200

Capacidades usadas:

- Clasificación de tráfico por protocolo (**NBAR**)
- Creación de clases (**class-map**)
- Aplicación de políticas (**policy-map**)
- Marcado de paquetes (**IPP y DSCP**)
- Aplicación en interfaces (**service-policy**)

3. Políticas de QoS

3.1 Justificación de selección de tráfico

- **HTTP:** Se eligió porque representa tráfico web común y fácil de generar en la red. Permite evaluar QoS en un escenario realista de uso cotidiano, donde múltiples usuarios acceden a servicios web.
- **SSH:** Se seleccionó por ser tráfico de administración remota. Aunque consume poco ancho de banda, requiere estabilidad y prioridad moderada para evitar interrupciones en la gestión de dispositivos.
- **ICMP:** Se utilizó porque es tráfico de diagnóstico (ping). Permite verificar conectividad y analizar cómo QoS trata tráfico que no es crítico, observando su comportamiento bajo políticas de priorización.

3.2 Clasificación (sin ACL)

```
class-map match-any CLASE-WEB  
match protocol http
```

```
class-map match-any CLASE-SSH  
match protocol ssh
```

```
class-map match-any CLASE-ICMP  
match protocol icmp
```

```
class-map match-any CLASE-VOIP  
match protocol rtp
```

3.3 Política aplicada

```
policy-map POLITICA-MAESTRA  
class CLASE-VOIP  
set dscp ef  
class CLASE-WEB  
set ip precedence 3  
class CLASE-SSH  
set dscp cs2
```

class CLASE-ICMP

set dscp af11

```
Class-map: CLASE-WEB (match-any)
  10 packets, 11618 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: protocol http
    10 packets, 11618 bytes
    5 minute rate 0 bps
  QoS Set
    precedence 3
    Packets marked 10

Class-map: CLASE-SSH (match-any)
  3166 packets, 351481 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: protocol ssh
    3166 packets, 351481 bytes
    5 minute rate 0 bps
  QoS Set
    dscp cs2
    Packets marked 3166

Class-map: CLASE-ICMP (match-any)
  24 packets, 2576 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: protocol icmp
    24 packets, 2576 bytes
    5 minute rate 0 bps
  QoS Set
    dscp af11
    Packets marked 24

Class-map: class-default (match-any)
  1684 packets, 136402 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: any
```

3.4 Aplicación en interfaces

interface Serial1/0

service-policy output POLITICA-MAESTRA

interface FastEthernet2/0

service-policy input POLITICA-MAESTRA

interface FastEthernet2/1

service-policy input POLITICA-MAESTRA

4. Marcado de paquetes

- **HTTP (WEB):** IP Precedence 3 → tráfico interactivo
- **SSH:** DSCP CS2 → administración
- **ICMP:** DSCP AF11 → diagnóstico
- **RTP (VoIP):** DSCP EF → tiempo real

5. Verificación

5.1 Comando de política

show policy-map

R1#show policy-map

Policy Map POLITICA-MAESTRA

Class CLASE-VOIP

set dscp ef

Class CLASE-WEB

set ip precedence 3

Class CLASE-SSH

set dscp cs2

Class CLASE-ICMP

set dscp af11

5.2 Verificación en interfaz

show policy-map interface FastEthernet2/1

R1#show policy-map interface fastEthernet 2/1

FastEthernet2/1

Service-policy input: POLITICA-MAESTRA

Class-map: CLASE-VOIP (match-any)

0 packets, 0 bytes

5 minute offered rate 0000 bps, drop rate 0000 bps

Match: protocol rtp

0 packets, 0 bytes

5 minute rate 0 bps

QoS Set

dscp ef

Packets marked 0

Class-map: CLASE-WEB (match-any)

10 packets, 11618 bytes

5 minute offered rate 0000 bps, drop rate 0000 bps

Match: protocol http

10 packets, 11618 bytes

5 minute rate 0 bps

QoS Set

precedence 3

Packets marked 10

Class-map: CLASE-SSH (match-any)

395 packets, 47851 bytes

5 minute offered rate 2000 bps, drop rate 0000 bps

Match: protocol ssh

395 packets, 47851 bytes

5 minute rate 2000 bps

QoS Set

dscp cs2

Packets marked 395

Class-map: CLASE-ICMP (match-any)

24 packets, 2576 bytes

5 minute offered rate 0000 bps, drop rate 0000 bps

Match: protocol icmp

24 packets, 2576 bytes

5 minute rate 0 bps

QoS Set

dscp af11

Packets marked 24

Class-map: class-default (match-any)

884 packets, 71602 bytes

5 minute offered rate 0000 bps, drop rate 0000 bps

Match: any

Descripción:

El tráfico coincide con la clasificación configurada y los paquetes muestran el marcado correcto según cada clase.

6. Captura de paquetes (Wireshark)

526	10.784379	192.168.11.20	192.168.11.130	TCP	66 40148 → 22 [ACK] Seq=7145 Ack=18601 Win=0 Len=0
527	10.784569	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)
528	10.785651	192.168.11.130	192.168.11.20	SSH	102 Server: Encrypted packet (len=36)
529	10.814881	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)
530	10.815009	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)

▶ Frame 526: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0	0000	0c 72 8f 7d 00 00 ca 01 7a
▶ Ethernet II, Src: ca:01:7a:a1:00:39 (ca:01:7a:a1:00:39), Dst: 0c:72:8f:7d:00:00 (0c:72:8f:7d:00:00)	0010	00 34 21 24 40 00 3f 06 82
▼ Internet Protocol Version 4, Src: 192.168.11.20, Dst: 192.168.11.130	0020	0b 82 9c d4 00 16 42 db 18
0100 = Version: 4	0030	01 70 ee 2b 00 00 01 01 08
.... 0101 = Header Length: 20 bytes (5)	0040	c9 d4
▼ Differentiated Services Field: 0x40 (DSCP: CS2, ECN: Not-ECT)		
0100 00.. = Differentiated Services Codepoint: Class Selector 2 (16)		
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)		
Total Length: 52		
Identification: 0x2124 (8484)		
▶ 010. = Flags: 0x2, Don't fragment		

Tráfico SSH

534	10.845500	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)
535	10.846545	192.168.11.130	192.168.11.20	TCP	66 22 → 40148 [ACK] Seq=18709 Ack=7325 Win
536	10.846901	192.168.11.130	192.168.11.20	SSH	138 Server: Encrypted packet (len=72)
537	10.885954	192.168.11.20	192.168.11.130	TCP	66 40148 → 22 [ACK] Seq=7325 Ack=18781 Win
538	10.886091	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)
539	10.887000	192.168.11.130	192.168.11.20	SSH	102 Server: Encrypted packet (len=36)
540	10.916647	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)
541	10.916838	192.168.11.20	192.168.11.130	SSH	102 Client: Encrypted packet (len=36)

▶ Frame 534: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface 0 ▶ Ethernet II, Src: ca:01:7a:a1:00:39 (ca:01:7a:a1:00:39), Dst: 0c:72:8f:7d:00:00 (0c:72:8f:7d:00:00) ▶ Internet Protocol Version 4, Src: 192.168.11.20, Dst: 192.168.11.130 0100 = Version: 4 0101 = Header Length: 20 bytes (5) ▼ Differentiated Services Field: 0x40 (DSCP: CS2, ECN: Not-ECT) 0100 00.. = Differentiated Services Codepoint: Class Selector 2 (16) 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0) Total Length: 88 Identification: 0x2129 (8489)	<pre> 0000 0c 72 8f 7d 00 00 ca 01 7a 0010 00 58 21 29 40 00 3f 06 82 0020 0b 82 9c d4 00 16 42 db 19 0030 01 70 bf 33 00 00 01 01 08 0040 ca 1b d5 4f 61 45 6a a5 8b 0050 d6 e5 55 e5 d2 fd cf 9d 41 0060 41 1d 44 37 62 c0 </pre>
---	---

7. Comprobación de Resultados:

- SSH de un cliente a otro:

```

user@debian:~$ ssh user@192.168.11.131
user@192.168.11.131's password:
Permission denied, please try again.
user@192.168.11.131's password:
Linux debian 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 29 14:38:03 2026 from 192.168.11.130
user@debian:~$ ls
prueba.txt
user@debian:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:ab:1b:16:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx0cab1b160000
    inet 192.168.11.131/25 brd 192.168.11.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::eab:1b16:0000:0000/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```


- Servidor Web Activo:

```

user@debian:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:72:8f:7d:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx0c728f7d0000
    inet 192.168.11.130/25 brd 192.168.11.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::c72:8fff:fe7d::4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
user@debian:~$ sudo systemctl status apache
Unit apache.service could not be found.
user@debian:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 13:20:46 CST; 1h 27min ago
     Invocation: d11d225bfff6400388d496a92ccc997a
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 543 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Process: 684 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 580 (apache2)
      Tasks: 55 (limit: 9470)
     Memory: 8.3M (peak: 9.9M)
        CPU: 848ms
    CGroup: /system.slice/apache2.service
            └─580 /usr/sbin/apache2 -k start
              └─699 /usr/sbin/apache2 -k start
                └─700 /usr/sbin/apache2 -k start

abr 29 13:20:46 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
abr 29 13:20:46 debian apachectl[575]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
abr 29 13:20:46 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
abr 29 13:22:01 debian systemd[1]: Reloading apache2.service - The Apache HTTP Server...
abr 29 13:22:01 debian apachectl[696]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
abr 29 13:22:01 debian systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
user@debian:~$

```

- Petición al Servidor Web:

```

</div>
Document Roots
</div>
<div class="content_section_text">
  <p>
    By default, Debian does not allow access through the web browser to
    <em>any</em> file apart of those located in <tt>/var/www</tt>,
    <a href="https://httpd.apache.org/docs/2.4/mod/mod_userdir.html" rel="nofollow">public_html</a>
    directories (when enabled) and <tt>/usr/share</tt> (for web
    applications). If your site is using a web document root
    located elsewhere (such as in <tt>/srv</tt>) you may need to whitelist your
    document root directory in <tt>/etc/apache2/apache2.conf</tt>.
  </p>
  <p>
    The default Debian document root is <tt>/var/www/html</tt>. You
    can make your own virtual hosts under /var/www. This is different
    to previous releases which provides better security out of the box.
  </p>
</div>
<div class="section_header">
  <div id="bugs"></div>
  Reporting Problems
</div>
<div class="content_section_text">
  <p>
    Please use the <tt>reportbug</tt> tool to report bugs in the
    Apache2 package with Debian. However, check <a
    href="https://bugs.debian.org/cgi-bin/pkgreport.cgi?ordering=normal;archive=0;src=apache2;repeatmerged=0"
    rel="nofollow">existing bug reports</a> before reporting a new bug.
  </p>
  <p>
    Please report bugs specific to modules (such as PHP and others)
    to respective packages, not to the web server itself.
  </p>
</div>
</div>
</div>
<div class="validator">
</div>
</body>
</html>
user@debian:~$ curl 192.168.11.130 _

```


- **Comprobación de Políticas:**

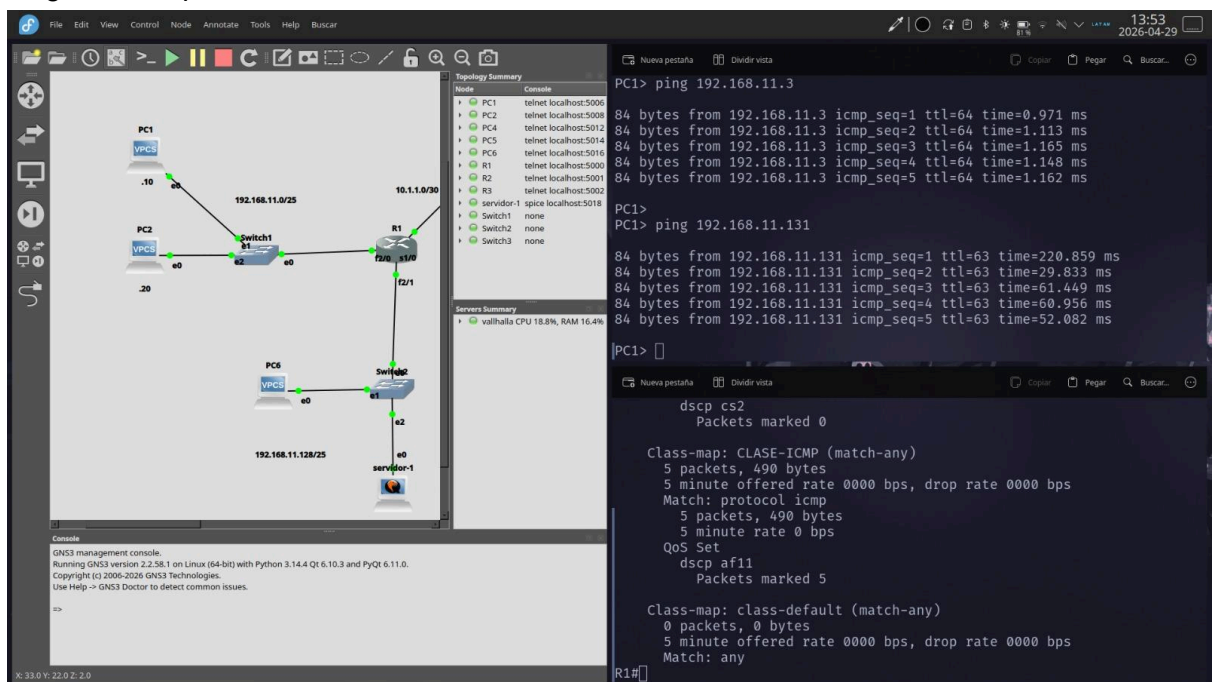
```
PC5> show ip

NAME       : PC5[1]
IP/MASK    : 172.16.12.20/24
GATEWAY    : 172.16.12.1
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10042
RHOST:PORT : 127.0.0.1:10043
MTU        : 1500

PC5> ping 192.168.11.20

192.168.11.20 icmp_seq=1 timeout
84 bytes from 192.168.11.20 icmp_seq=2 ttl=61 time=114.071 ms
84 bytes from 192.168.11.20 icmp_seq=3 ttl=61 time=121.575 ms
84 bytes from 192.168.11.20 icmp_seq=4 ttl=61 time=122.847 ms
84 bytes from 192.168.11.20 icmp_seq=5 ttl=61 time=110.695 ms
```

- **Ping entre dispositivos:**



The screenshot displays the GNS3 interface with a network topology on the left and terminal outputs on the right. The topology shows PC1 and PC2 connected to Switch1, which is connected to R1. R1 is connected to Switch2, which is connected to PC5 and a server. The terminal outputs show ping results between PC1 and 192.168.11.3, and between PC1 and 192.168.11.131. The console output shows the GNS3 management console running on Linux.

```
PC1> ping 192.168.11.3
84 bytes from 192.168.11.3 icmp_seq=1 ttl=64 time=0.971 ms
84 bytes from 192.168.11.3 icmp_seq=2 ttl=64 time=1.113 ms
84 bytes from 192.168.11.3 icmp_seq=3 ttl=64 time=1.165 ms
84 bytes from 192.168.11.3 icmp_seq=4 ttl=64 time=1.148 ms
84 bytes from 192.168.11.3 icmp_seq=5 ttl=64 time=1.162 ms

PC1> ping 192.168.11.131
84 bytes from 192.168.11.131 icmp_seq=1 ttl=63 time=220.859 ms
84 bytes from 192.168.11.131 icmp_seq=2 ttl=63 time=29.833 ms
84 bytes from 192.168.11.131 icmp_seq=3 ttl=63 time=61.449 ms
84 bytes from 192.168.11.131 icmp_seq=4 ttl=63 time=60.956 ms
84 bytes from 192.168.11.131 icmp_seq=5 ttl=63 time=52.082 ms

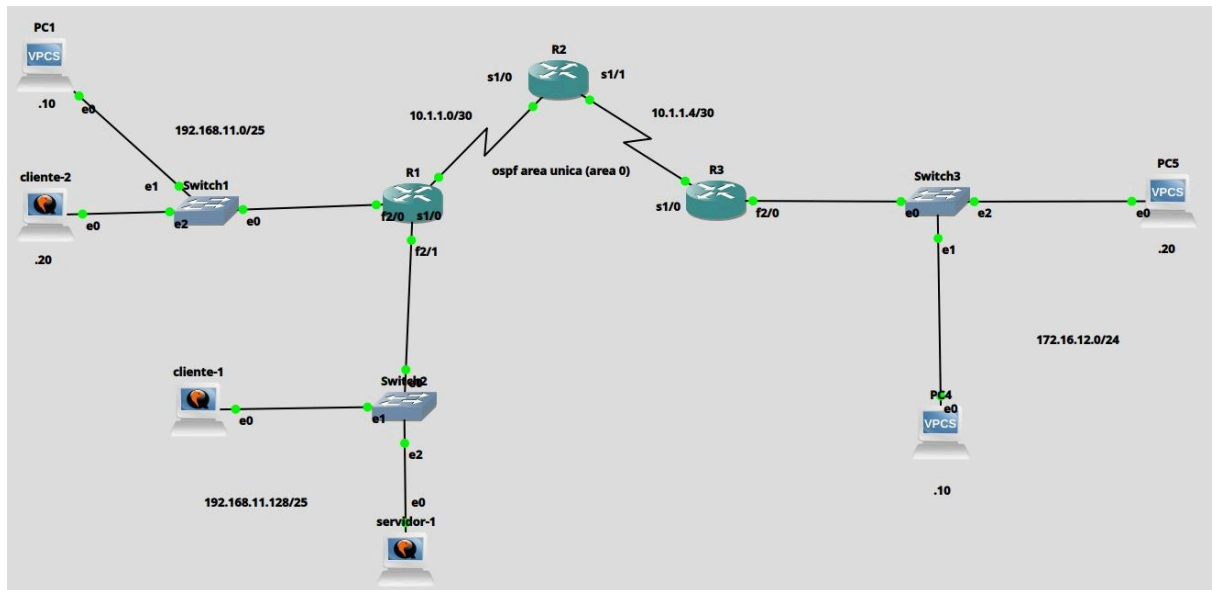
PC1>

dscp cs2
Packets marked 0

Class-map: CLASE-ICMP (match-any)
  5 packets, 400 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
Match: protocol icmp
  5 packets, 400 bytes
  5 minute rate 0 bps
QoS Set
  dscp af11
  Packets marked 5

Class-map: class-default (match-any)
  0 packets, 0 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
Match: any
```

- **Topología Final:**



8. Conclusión

Las políticas de QoS se implementaron correctamente utilizando clasificación por protocolo sin ACL.

La marcación de paquetes con IP Precedence y DSCP se verificó mediante comandos y capturas en Wireshark, confirmando su correcto funcionamiento.